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General Knowledge

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$$G(s) = C(Is - A)^{-1}B + D \quad AQ = Q\Lambda \quad AV = VJ$$

- $\det(A) \neq 0$, invertible, full rank, lin. ind. cols/rows, nullity=0, non-singular
- Zero-state is "synonym" for TF thinking
- D, U, and L matrices show eigs on diag
- A J Matrix has 1 block for each grade-1 eigenvector. Ex: 3x3 w/ multiplicity 3 results in one J block as the entire matrix
- Eigvector order in V and J match
- nullity($A - \lambda_i I$) = # of J-blocks for λ_i
- Basis is lin. ind. cols of A , null space is remaining cols of A
- Z transform wants func. of z^{-1} not z
- A square matrix satisfies it's own $\Delta(\lambda)$
- SS models are equiv. if same eigenvalues. Zero-state equiv. if same TF (matrix)

$$\overbrace{y_3(t) = \alpha y_1(t)}^{\text{Homogeneity}} \quad \text{if} \quad \begin{cases} x_3(0) = \alpha x_1(0) \\ u_3(t) = \alpha u_1(t) \end{cases}$$

$$\overbrace{y_3(t) = y_1(t) + y_2(t)}^{\text{Additivity}} \quad \text{if} \quad \begin{cases} x_3(0) = x_1(0) + x_2(0) \\ u_3(t) = u_1(t) + u_2(t) \end{cases}$$

$$\hat{A}_{\text{comp.}} = C^{-1}AC \quad \text{follow through in sys.}$$

$$\begin{bmatrix} -\alpha_1 & -\alpha_2 & -\alpha_3 & -\alpha_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\Delta(\lambda) = \lambda^4 + \alpha_1 \lambda^3 + \alpha_2 \lambda^2 + \alpha_3 \lambda + \alpha_4$$

$$(A - I\lambda)v_i = q_i \iff v_i \text{ is gen. eigvec of } A$$

$$y[k+2] + 5y[k+1] + 10y[k] = 7u[k]$$

$$z^2Y[z] + 5zY[z] + 10Y[z] = 7U[z]$$

General Solutions

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$$

$$x[k] = A^k x[0] \quad \text{for y, plug x into y eq.}$$

$$x(t) = e^{Q\Lambda Q^{-1}t}x(0) = Qe^{\Lambda t}Q^{-1}x(0)$$

Functions of Matrices

$$e^{At} = \begin{bmatrix} e^{\lambda t} & te^{\lambda t} & \frac{t^2}{2!}e^{\lambda t} & \frac{t^3}{3!}e^{\lambda t} \\ 0 & e^{\lambda t} & te^{\lambda t} & \frac{t^2}{2!}e^{\lambda t} \\ 0 & 0 & e^{\lambda t} & te^{\lambda t} \\ 0 & 0 & 0 & e^{\lambda t} \end{bmatrix}$$

(applies to J blocks)

$$e^A = 1 + A^2/2! + A^3/3! + A^4/4! + \dots$$

$$\sin(A) = A - A^3/3! + A^5/5! - A^7/7! + \dots$$

$$\cos(A) = 1 - A^2/2! + A^4/4! - A^6/6! + \dots$$

$$f(\hat{A}) = Q^{-1}f(A)Q \quad f(A) = Q^{-1}f(\hat{A})Q$$

$$\text{if } A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \rightarrow f(A) = \begin{bmatrix} f(A_1) & 0 \\ 0 & f(A_2) \end{bmatrix}$$

$$J^k = \begin{bmatrix} \lambda^k & k\lambda^{k-1} & \frac{k(k-1)}{2!}\lambda^{k-2} \\ 0 & \lambda^k & k\lambda^{k-1} \\ 0 & 0 & \lambda^k \end{bmatrix}$$

(single block)

Polynomials

$\Delta(\lambda)$ has all eigvalues. $\psi(\lambda)$ only includes the single largest J block for each eigvalue Ex:

$$\begin{bmatrix} \lambda_1 & 1 & 0 & 0 \\ 0 & \lambda_1 & 1 & 0 \\ 0 & 0 & \lambda_1 & 0 \\ 0 & 0 & 0 & \lambda_2 \end{bmatrix} \begin{cases} \Delta(\lambda) = (\lambda - \lambda_1)^3(\lambda - \lambda_2) \\ \psi(\lambda) = (\lambda - \lambda_1)^3(\lambda - \lambda_2) \end{cases}$$

$$\begin{bmatrix} \lambda_1 & 1 & 0 & 0 \\ 0 & \lambda_1 & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 \\ 0 & 0 & 0 & \lambda_1 \end{bmatrix} \begin{cases} \Delta(\lambda) = (\lambda - \lambda_1)^4 \\ \psi(\lambda) = (\lambda - \lambda_1)^2 \end{cases}$$

Realizations (method two) (for i inputs)

$$G(s) = \frac{\begin{bmatrix} \beta_{11}s^3 & \beta_{12}s^2 & \beta_{13}s & \beta_{14} \\ \beta_{21}s^3 & \beta_{22}s^2 & \beta_{23}s & \beta_{24} \end{bmatrix}}{s^4 + \alpha_1s^3 + \alpha_2s^2 + \alpha_3s + \alpha_4} + \overbrace{\begin{bmatrix} d_1 \\ d_2 \end{bmatrix}}^{G(\infty)}$$

$$A = \begin{bmatrix} -\alpha_1 & -\alpha_2 & -\alpha_3 & -\alpha_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$B_i = [1 \quad 0]^T \quad D_i = [d_1 \quad d_2]^T$$

$$C_i = \begin{bmatrix} \beta_{11} & \beta_{12} & \beta_{13} & \beta_{14} \\ \beta_{21} & \beta_{22} & \beta_{23} & \beta_{24} \end{bmatrix}$$

Then

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1 & 0 \\ 0 & b_2 \end{bmatrix} u$$

$$y = [C_1 \quad C_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [d_1 \quad d_2] u$$

Stability

BIBO: zero-state response $\rightarrow G(0) * u(t)$. Stable iff every pole in OLHP/open unit circle
AS: all $Re(\lambda_i) < 0$ for C, or $|\lambda_i| < 1$ for D.
And $x_i(\infty) \rightarrow 0 \quad \forall i$

MS: Allow $Re(\lambda_i) = 0$ for C, or $|\lambda_i| = 1$ for D, only for simple roots (J-blocks sized 1)

Controllable If

- $\rho(C) = \rho([B \quad AB \quad \dots \quad A^{n-1}B]) = n$
- $\rho([A - \lambda I \quad B]) = n \quad \forall \lambda_i$
- Rows(B) aligned with bottom of each J-block are lin ind if repeated, else nonzero

Observable If

- $\rho(O) = \rho[C \quad CA \quad \dots \quad CA^{n-1}]^T = n$
- $\rho([A - \lambda I \quad C]^T) = n \quad \forall \lambda_i$
- Cols(C) aligned with left of each J-block are lin ind if repeated, else nonzero

Kalman Decomposition

if $\rho(C) = n_1 < n$:

form $P^{-1} = [q_1 \quad \dots \quad q_{n_1} \quad \dots q_n]$ with n_1 lin ind cols of C , and rest to make invertible, then $P^{-1}\hat{x} = x$ and

$$\begin{bmatrix} \hat{x}_C \\ \hat{x}_{\bar{C}} \end{bmatrix} = \begin{bmatrix} \hat{A}_C & \hat{A}_{12} \\ 0 & \hat{A}_{\bar{C}} \end{bmatrix} \begin{bmatrix} \hat{x}_C \\ \hat{x}_{\bar{C}} \end{bmatrix} + \begin{bmatrix} \hat{B}_C \\ 0 \end{bmatrix} u$$

$$y = [\hat{C}_C \quad \hat{C}_{\bar{C}}] \begin{bmatrix} \hat{x}_C \\ \hat{x}_{\bar{C}} \end{bmatrix} + Du$$

if $\rho(\mathcal{O}) = n_2 < n$:

form $P = [P_1 \quad \dots \quad P_{n_2} \quad \dots P_n]^T$ with n_2 lin
ind rows of $\mathcal{O}$, and rest to make invertible,
then $P^{-1}\hat{x} = x$ and

$$\begin{bmatrix} \hat{x}_O \\ \hat{x}_{\overline{O}} \end{bmatrix} = \begin{bmatrix} \hat{A}_O & 0 \\ \hat{A}_{21} & \hat{A}_{\overline{O}} \end{bmatrix} \begin{bmatrix} \hat{x}_O \\ \hat{x}_{\overline{O}} \end{bmatrix} + \begin{bmatrix} \hat{B}_O \\ \hat{B}_{\overline{O}} \end{bmatrix} u$$

$$y = [\hat{C}_O \quad 0] \begin{bmatrix} \hat{x}_O \\ \hat{x}_{\overline{O}} \end{bmatrix} + Du$$

Controllability After Sampling

If (A, B) is controllable, a sufficient
condition for (A_d, B_d) controllability is:

if $Re(\lambda_i) - Re(\lambda_j) = 0 \quad \lambda_i, \lambda_j \in A$, then

pick $T : |Im(\lambda_i)| - |Im(\lambda_j)| \neq \frac{2\pi m}{T} \quad m \in \mathcal{N}_1$

CL State Feedback

$$u = r - k \otimes x \quad \dot{x} = (A - Bk)x + Br$$

$(A - Bk, B)$ is controllable iff (A, B) is

$$k_i = \hat{\alpha}_i - \alpha_i = \text{des} - \text{current}$$

$$x_{ss} = -(A - Bk)^{-1}Br \quad y_{ss} = Cx_{ss}$$

1. Controllable?

2. Find $A_{CL} = (A - B \otimes k)$ in terms of k_i

3. $\Delta_{CL}(s) = \det(\lambda I - A_{CL}) \rightarrow \alpha_i$

4. Determine α_i from $\Delta_{des}(s)$

5. $g_{dc} = C(A - Bk)^{-1}B + D \quad r = y/g_{dc}$

Stabilization

$$\begin{bmatrix} \hat{x}_C \\ \hat{x}_{\overline{C}} \end{bmatrix} = \begin{bmatrix} \hat{A}_C - \hat{B}_C \hat{k}_C & \hat{A}_{12} - \hat{B}_C \hat{k}_{\overline{C}} \\ 0 & \hat{A}_{\overline{C}} \end{bmatrix} \begin{bmatrix} \hat{x}_C \\ \hat{x}_{\overline{C}} \end{bmatrix} + \begin{bmatrix} \hat{B}_C \\ 0 \end{bmatrix} r$$

if $u = r - [k_C \quad k_{\overline{C}}] \otimes [x_C \quad x_{\overline{C}}]^T$

Stabilizable: if uncontrollable states are al-
ready stable

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$$A^{-1} = \frac{A_{co}^T}{\det(A)} \quad ||x||_k \triangleq \left(\sum_{i=1}^n |x_i|^k \right)^{1/k}$$

Method ONE:

$$D = G(\infty)$$

$$G_{sp}(s) = \frac{1}{D(s)} [N_1 s^{n-1} \quad N_2 s^{n-2} \quad \dots \quad N_n]$$

$$A = \begin{bmatrix} -\alpha_1 I & -\alpha_2 I & \dots & -\alpha_{n-1} I & -\alpha_n I \\ I & 0 & \dots & 0 & 0 \\ 0 & I & \dots & 0 & 0 \\ & & \ddots & & \\ 0 & 0 & \dots & I & 0 \end{bmatrix}$$

$$B = [I \quad 0 \quad 0 \quad \dots \quad 0]^T \quad C = [N_1 + \dots + N_n]$$

Will work, but not in fewest states.

Method TWO: $D = G(\infty)$

Generate SS models column by column

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1 & 0 \\ 0 & b_2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$y = [C_1 \quad C_2] \vec{x} + G(\infty) \vec{u}$$

Will work, in fewest possible states.

LQR

$$\min(J) = \int_0^\infty (x^T(\tau)Qx(\tau) + u^T(\tau)Ru(\tau))d\tau$$

Q must be +ve semi-definite. $x^T(t)Qx(t) \geq 0 \forall x(t) \neq 0$. Diags must be positive OR zero. If any off-diag. causes any $\lambda = 0$, not semi-definite.

R must be positive definite. $u^T(t)Ru(t) > 0 \forall u(t) \neq 0$. Diags must be greater than zero.

$$u = -Kx \quad K = R^{-1}B^TP \text{ where } P \text{ solves:}$$

$$ARE: \quad A^TP + PA - PBR^{-1}B^TP + Q = 0$$

LQR is iterative. Pick Q and R , run sim with hard ic's and tweak to achieve desired behavior of system.

Full Kalman Decomposition

$$\begin{bmatrix} \hat{x}_{CO} \\ \hat{x}_{C\bar{O}} \\ \hat{x}_{\bar{C}O} \\ \hat{x}_{\bar{C}\bar{O}} \end{bmatrix} = \begin{bmatrix} \hat{A}_{CO} & 0 & \hat{A}_{13} & 0 \\ \hat{A}_{21} & \hat{A}_{C\bar{O}} & \hat{A}_{23} & \hat{A}_{24} \\ 0 & 0 & \hat{A}_{\bar{C}O} & 0 \\ 0 & 0 & \hat{A}_{43} & \hat{A}_{\bar{C}\bar{O}} \end{bmatrix} \begin{bmatrix} \hat{x}_{CO} \\ \hat{x}_{C\bar{O}} \\ \hat{x}_{\bar{C}O} \\ \hat{x}_{\bar{C}\bar{O}} \end{bmatrix} + [\hat{B}_{CO} \quad \hat{B}_{C\bar{O}} \quad 0 \quad 0]^T u$$

$$y = [\hat{C}_{CO} \quad 0 \quad \hat{C}_{\bar{C}O} \quad 0] \begin{bmatrix} \hat{x}_{CO} \\ \hat{x}_{C\bar{O}} \\ \hat{x}_{\bar{C}O} \\ \hat{x}_{\bar{C}\bar{O}} \end{bmatrix} + Du$$

Kalman Filter

Incorporate system and sensor noise as a benefit

1. The discrete system model

$$\begin{aligned} x_{t+1} &= Ax_t + Bu_t + m & m &\sim \mathcal{N}(0, M) \\ z_{t+1} &= Cx_t + n & n &\sim \mathcal{N}(0, N) \end{aligned}$$

2. Incorporate the past system behavior

$$x_{t-1} \sim \mathcal{N}(\hat{x}_{t-1}, \varepsilon_{t-1})$$

3. Do the process update (in code)

$$\begin{aligned} \hat{x}'_t &= A\hat{x}_{t-1} + Bu_{t-1} \\ \varepsilon'_t &= A\varepsilon_{t-1}A^T + m \end{aligned}$$

4. Do the measurement update (in code)

$$K_t = \varepsilon'_t C^T (C \varepsilon'_t C^T + n)^{-1}$$

$$\hat{x}_t = \underbrace{\hat{x}'_t}_{\text{w/o sensors}} + \underbrace{K_t (z_t - C\hat{x}'_t)}_{\text{w/ sensors innovation}}$$

$$\varepsilon_t = \varepsilon'_t - K_t C \varepsilon'_t \quad // \text{ Update uncert.}$$

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